BUS4498 Mini AI Build-a-thon | Option 1

FlexRoutine Planner - Maintain habits when schedules change

Problem
A student wants to keep important habits, but classes, shifts, and unexpected events change the time available.

Useful endpoint
A realistic adjusted routine the student can follow immediately, with full or shortened habits, skipped habits, total minutes, and a supportive explanation.

Trigger to endpoint workflow

1. Trigger

Student enters the schedule change and available minutes.

2. Collect

App gathers habits, priority labels, full minutes, minimum minutes, and details.

3. Plan

Claude interprets the change, protects priorities, and proposes full or shortened blocks.

4. Validate

Program checks totals and runs the Level 3 correction loop if needed.

5. Endpoint

Dashboard shows the adjusted routine, skipped habits, totals, and explanation.

AI and validation tasks

- **Interpret:** natural-language schedule changes.
- **Prioritize:** essential, important, optional labels.
- **Adapt:** use full habits when they fit; otherwise use minimum versions.
- **Explain:** give short, supportive reasons.
- **Level 3:** detect an over-budget draft, shorten or remove lower-priority items, recalculate, and recheck before display.

Models, tools, and Skill choice

- **Model:** Claude through the Anthropic Messages API; useful for natural-language interpretation and structured JSON.
- **App:** local Streamlit dashboard; fast to demonstrate and no public deployment needed.
- **Data:** JSON between the app and model; easy to validate and download.
- **Skill:** FlexRoutine Planner standardizes priority rules, fallback behavior, supportive language, and output format.
- **Key:** ANTHROPIC_API_KEY in local .env; .env is gitignored and never displayed.

Potential Level 3 task - self-correction loop
The agent drafts a plan. The program checks whether total_minutes is within available_minutes. If not, the agent revises lower-priority items, shortens or skips them, recalculates totals, and the app rechecks the result before presenting it.

Demo example
25 minutes available | Study = essential (30/10 min) | Exercise = important (30/10 min) | Meditation = optional (10/2 min)

Claude proposes a feasible shortened routine, explains the priority decisions, and identifies anything skipped.

Team prep checklist
- Confirm teammate and both speaking roles
- Confirm Claude/API choice and local .env key
- Test full-time, limited-time, and zero-time cases
- Rehearse the 3-minute local showcase
- Explain the Skill and Level 3 correction loop

Files: app.py | skills/flexroutine-planner/SKILL.md | agents/openai.yaml (Codex skill metadata) | .env (local secret) | docs/level-3-correction-loop.md